

A 26-VERTEX COUNTEREXAMPLE TO THE LITERAL WEAK-INEQUALITY FORM OF AIM CONJECTURE 6.23

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. We give an explicit finite loopless simple digraph with nonnegative rational arc weights such that every vertex has total inweight and total outweight at least one, while every simple directed cycle has weight strictly less than one. The construction has 26 vertices and 74 arcs. It has exactly 40 simple directed cycles—twenty digons and twenty triangles—and every one has weight $19/20$. This disproves the literal weak-inequality formulation printed as AIM Conjecture 6.23. The example has two strongly connected components and does not satisfy equality of all local weights, so it does not address strongly connected or equality-constrained formulations.

1. INTRODUCTION

Let G be a finite digraph and let $p : E(G) \rightarrow [0, 1]$ assign a weight to every arc. For $u \in V(G)$ define

$$p^+(u) = \sum_{uv \in E(G)} p(uv), \quad p^-(u) = \sum_{vu \in E(G)} p(vu).$$

The statement printed as Conjecture 6.23 in the AIM survey on the Caccetta–Häggkvist conjecture is the following [3, p. 9].

If $p : E(G) \rightarrow [0, 1]$ and $p^-(u), p^+(u) \geq 1$ for every $u \in V(G)$, then G contains a directed cycle of total weight at least 1.

Here and below, a directed cycle is simple: its vertices are distinct except for the repeated initial and terminal vertex. We prove the following.

Theorem 1.1. *There exists a finite loopless simple weighted digraph (G, p) such that*

$$p^-(v) \geq 1 \quad \text{and} \quad p^+(v) \geq 1 \quad (v \in V(G)),$$

but

$$\max \left\{ \sum_{e \in C} p(e) : C \text{ is a simple directed cycle of } G \right\} = \frac{19}{20} < 1.$$

One such graph has 26 vertices, 74 arcs, and exactly 40 simple directed cycles.

The proof is elementary. The graph consists of two isomorphic 13-vertex pieces with opposite orientations, joined by ten arcs all directed from one piece to the other. The one-way join raises the deficient local weights without creating any new directed cycle.

Remark 1.2 (Scope). The graph in [Theorem 1.1](#) is not strongly connected. Moreover, some local inweights and outweights are strictly greater than 1. Consequently the example does not refute a version requiring strong connectivity, nor a version requiring $p^-(v) = p^+(v) = 1$ at every vertex. These distinctions are essential and are retained throughout the paper.

2. THE CONSTRUCTION

Let $I = \{0, 1\}$ and $J = \{1, 2, 3, 4, 5\}$. For each sign $\sigma \in \{-, +\}$ define

$$V_\sigma = \{r_\sigma, a_{0,\sigma}, a_{1,\sigma}\} \cup \{b_{ij,\sigma} : i \in I, j \in J\}.$$

Set $V(G) = V_- \sqcup V_+$. Thus $|V(G)| = 26$.

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Inside V_+ , for every $i \in I$ and $j \in J$, add the arcs

$$\begin{aligned} r_+ &\longrightarrow a_{i,+}, & p &= \frac{1}{2}, \\ a_{i,+} &\longrightarrow b_{ij,+}, & p &= \frac{1}{5}, \\ b_{ij,+} &\longrightarrow r_+, & p &= \frac{1}{4}, \\ b_{ij,+} &\longrightarrow a_{i,+}, & p &= \frac{3}{4}. \end{aligned} \tag{1}$$

Inside V_- , reverse every arc in (1) and retain its weight. Explicitly, add

$$\begin{aligned} a_{i,-} &\longrightarrow r_-, & p &= \frac{1}{2}, \\ b_{ij,-} &\longrightarrow a_{i,-}, & p &= \frac{1}{5}, \\ r_- &\longrightarrow b_{ij,-}, & p &= \frac{1}{4}, \\ a_{i,-} &\longrightarrow b_{ij,-}, & p &= \frac{3}{4}. \end{aligned} \tag{2}$$

Finally, add the ten cross arcs

$$b_{ij,-} \longrightarrow b_{ij,+} \quad \text{with weight } \frac{4}{5} \quad (i \in I, j \in J). \tag{3}$$

There are 32 arcs in each induced subgraph $G[V_\sigma]$ and ten cross arcs, hence

$$|E(G)| = 32 + 32 + 10 = 74.$$

Every weight belongs to $[0, 1]$. The displayed list contains neither a loop nor two arcs with the same ordered pair of endpoints, so G is loopless and simple.

3. VERIFICATION OF THE LOCAL WEIGHT CONDITIONS

All vertices of a fixed displayed type have the same inweight and outweight. Directly from (1)–(3) one obtains Table 1.

TABLE 1. Local inweights and outweights.

Vertex type	Number of vertices	$p^-(v)$	$p^+(v)$
r_-	1	1	$5/2$
$a_{i,-}$	2	1	$17/4$
$b_{ij,-}$	10	1	1
r_+	1	$5/2$	1
$a_{i,+}$	2	$17/4$	1
$b_{ij,+}$	10	1	1

For completeness, the entries follow as follows. The vertex r_- receives two arcs of weight $1/2$ and sends ten arcs of weight $1/4$. Each $a_{i,-}$ receives five arcs of weight $1/5$ and sends one arc of weight $1/2$ together with five arcs of weight $3/4$. Each $b_{ij,-}$ receives weights $1/4$ and $3/4$ and sends weights $1/5$ and $4/5$. The plus-side calculations are the reversed counterparts, with the cross arcs supplying the missing $4/5$ at the leaves.

It follows immediately that

$$\min_{v \in V(G)} p^-(v) = 1, \quad \min_{v \in V(G)} p^+(v) = 1. \tag{4}$$

4. CLASSIFICATION OF THE DIRECTED CYCLES

Lemma 4.1. *The sets V_- and V_+ induce the two strongly connected components of G . Every arc between the components is directed from V_- to V_+ .*

Proof. In $G[V_+]$, every $a_{i,+}$ is reachable from r_+ , every $b_{ij,+}$ is reachable from $a_{i,+}$, and every $b_{ij,+}$ has an arc back to r_+ . Thus $G[V_+]$ is strongly connected. Reversing all internal arcs proves the same assertion for $G[V_-]$. The final claim is exactly (3), and there is no arc from V_+ to V_- . Hence the two sets are distinct strongly connected components. $\square$

Lemma 4.2. *For each $\sigma \in \{-, +\}$, $i \in I$, and $j \in J$, the component $G[V_\sigma]$ has exactly the following two cycles involving $b_{ij,\sigma}$:*

- (i) the digon $a_{i,\sigma} \leftrightarrow b_{ij,\sigma}$;
- (ii) the triangle on $\{r_\sigma, a_{i,\sigma}, b_{ij,\sigma}\}$, with the orientation prescribed by (1) or (2).

There are no other simple directed cycles in G .

Proof. By Lemma 4.1, no directed cycle can use a cross arc. It therefore suffices to inspect $G[V_+]$; the minus component is obtained by reversing every arc.

Every outgoing arc of $a_{i,+}$ enters some $b_{ij,+}$. From that leaf there are exactly two internal choices. Returning to $a_{i,+}$ gives the digon in (i). Moving to r_+ forces the next vertex to be one of $a_{0,+}$ and $a_{1,+}$. Choosing $a_{i,+}$ closes the triangle in (ii). Choosing the other a -vertex forces a later repetition of either that a -vertex or r_+ before the initial vertex can be reached, which is incompatible with simplicity. This exhausts all possibilities. $\square$

For each sign there are $|I||J| = 10$ cycles of each type. Hence G has twenty digons and twenty triangles. Their respective weights are

$$\frac{3}{4} + \frac{1}{5} = \frac{19}{20}, \quad \frac{1}{2} + \frac{1}{5} + \frac{1}{4} = \frac{19}{20}.$$

Together with (4), this proves Theorem 1.1.

5. INDEPENDENT COMPUTATIONAL CHECK

The preceding proof is independent of computation. As a separate check, the graph was reconstructed from the four internal arc families and the cross-arc family using integer arithmetic at scale 20. The calculation checked all local weights, computed mutual reachability, and enumerated simple directed cycles by subset dynamic programming, with cycles identified modulo cyclic rotation.

The recorded output is

Quantity	Exact value
Vertices	26
Arcs	74
Strongly connected component sizes	13, 13
Cross-component arcs	10
Simple directed cycles	40
Cycle-length histogram	20 of length 2; 20 of length 3
Maximum cycle weight	19/20

An independent reviewer rebuilt the adjacency lists and used a separate depth-first enumeration. That computation returned the same vertex and arc counts, the same two strongly connected components, the same canonical cycle set, and the same maximum weight.

6. DISCUSSION OF THE FORMULATION

The construction exploits a precise difference between the literal weak inequalities and stronger formulations. Internally, both 13-vertex components have controlled cycles of weight 19/20. The leaves would have insufficient local weight on one side if the components were considered separately. The ten one-way cross arcs repair exactly those local deficits but, because the condensation digraph is acyclic, create no new cycle.

This mechanism is unavailable if the whole graph is required to be strongly connected. It also does not produce a counterexample when every local inweight and outweight is required to equal 1, since Table 1 contains the values 5/2 and 17/4. The present note therefore settles only the statement as printed in [3]. Related weighted-path and heavy-cycle results may be found in [1, 2].

The example is small, exact, and independently reproducible. No minimality assertion is made: the paper does not claim that 26 is the least possible number of vertices, nor that 74 is the least possible number of arcs.

APPENDIX A. CHECKLIST FOR AN INDEPENDENT REFEREE

The following checks suffice to verify the result without relying on the audit pipeline.

- (1) Reconstruct the 26 vertices and 74 arcs from (1)–(3).
- (2) Check that the graph is loopless and has no parallel arcs, and that all weights belong to $[0, 1]$.

- (3) Recompute the six rows of Table 1.
- (4) Verify that V_- and V_+ are the two strongly connected components and that all ten cross arcs point from V_- to V_+ .
- (5) Enumerate simple directed cycles up to cyclic rotation and confirm that only the two families in Lemma 4.2 occur.
- (6) Confirm the histogram $\{2 : 20, 3 : 20\}$ and the common weight $19/20$.
- (7) Compare the construction only with the literal weak-inequality statement, keeping equality and strong-connectivity variants separate.

REFERENCES

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